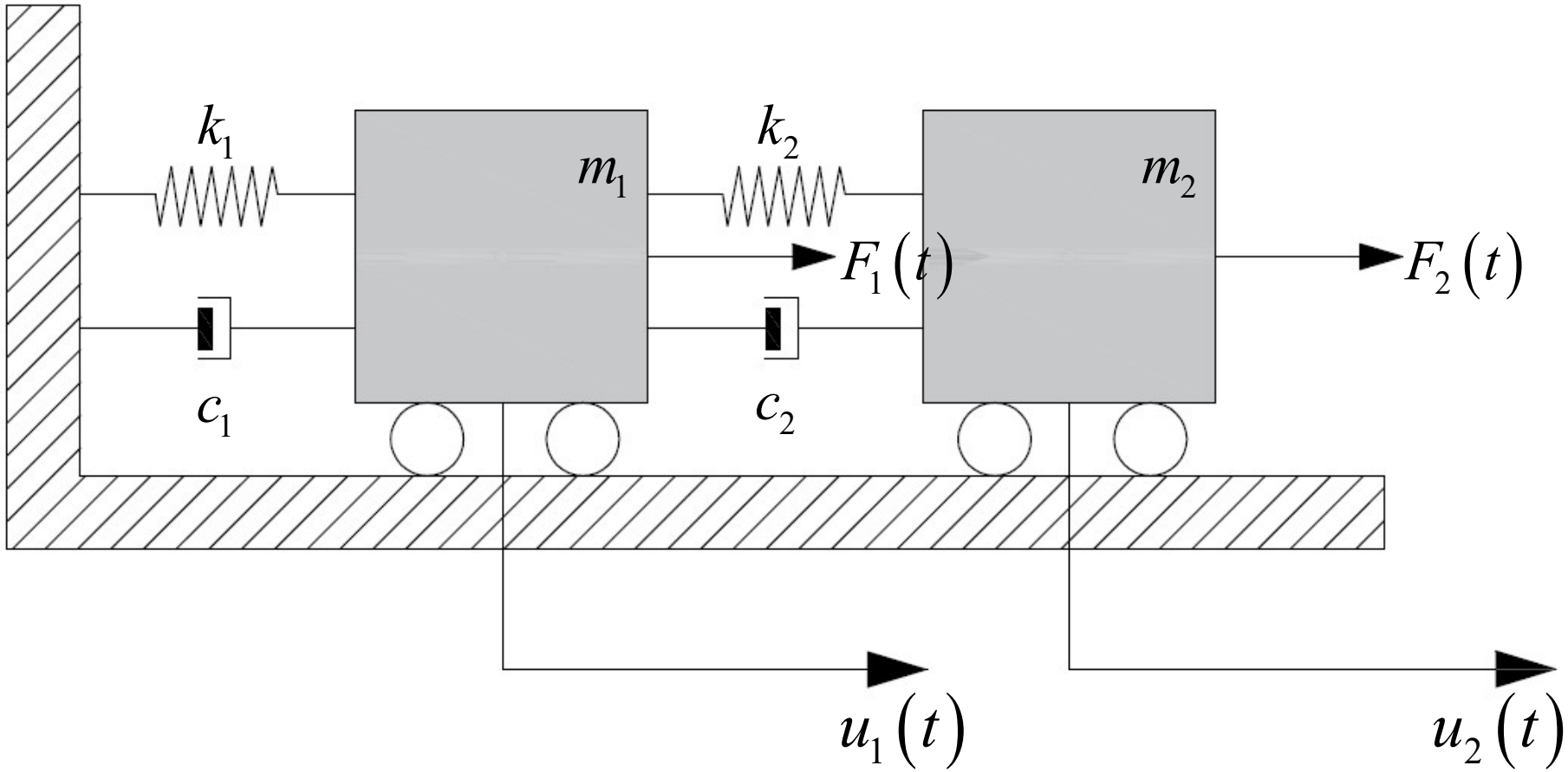
MDOF systems
Introduction: degrees of freedom and generalized coordinates

We now turn our attention to a generic discrete multi-degree of freedom (MDOF) system.
The motion of a generic MDOF system can be described by a finite number of degrees of freedom, collected into a vector $\mathbf{u}(t)$, whose components, in Dynamics, are all <u>independent</u> unknown functions of time.

A typical, although extremely simple, example of such a system is given in the Figure.



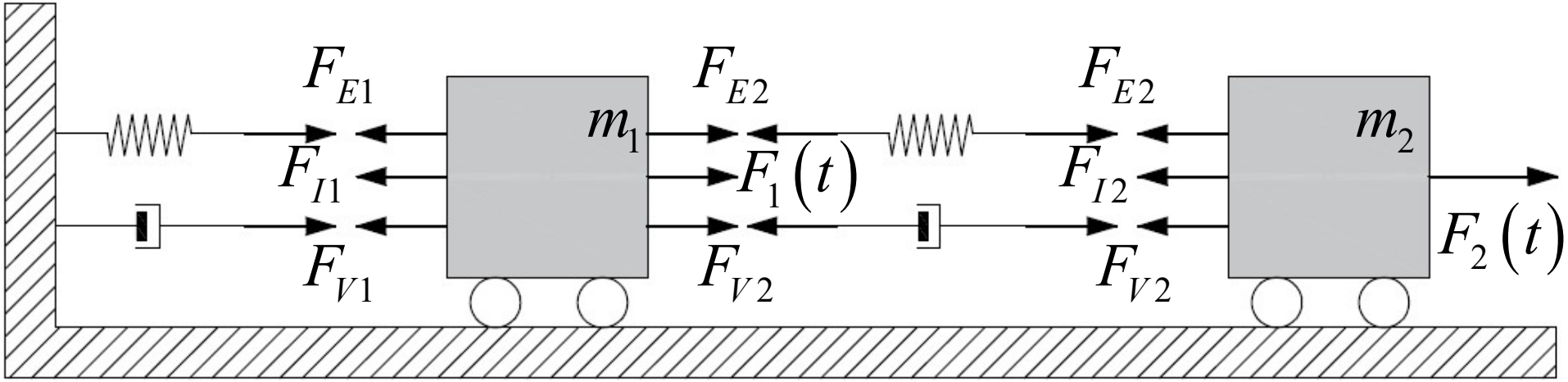
In this example, $u_1(t)$ and $u_2(t)$ are the absolute displacements – i.e., referred to the same fixed reference system – of the two masses.

This is a discrete problem with 2 degrees of freedom, since each mass can move by the independent quantity $u_1(t)$ and $u_2(t)$.

The equations governing the motion of the two masses can be easily written in terms of what we have already seen for the SDOF systems.

To this purpose, one must evidentiate all the forces acting on the two masses, and proceed for each mass as seen for the SDOF case.

For this elementary case, it is sufficient to put together, by inspection, the main basic equations, i.e., dynamic equilibrium, compatibility, and statics-kinematics laws.

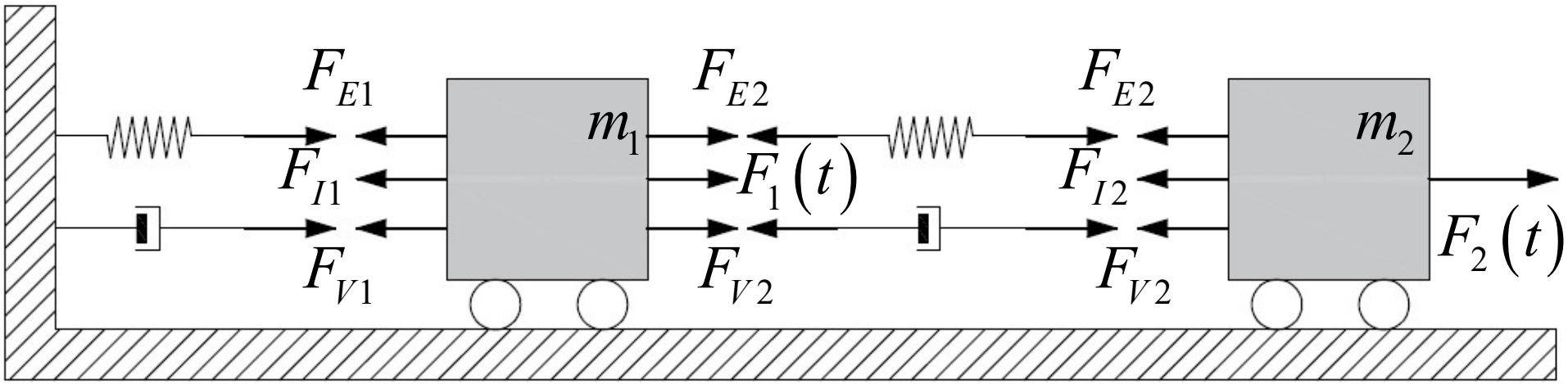


The <u>statics-kinematics laws</u> remain unaltered:

$$F_{Ii}(t) = m_i \ddot{u}_i(t);$$

$$F_{Vi}(t) = c_i \Delta \dot{u}_i(t);$$

$$F_{Ei}(t) = k_i \Delta u_i(t)$$

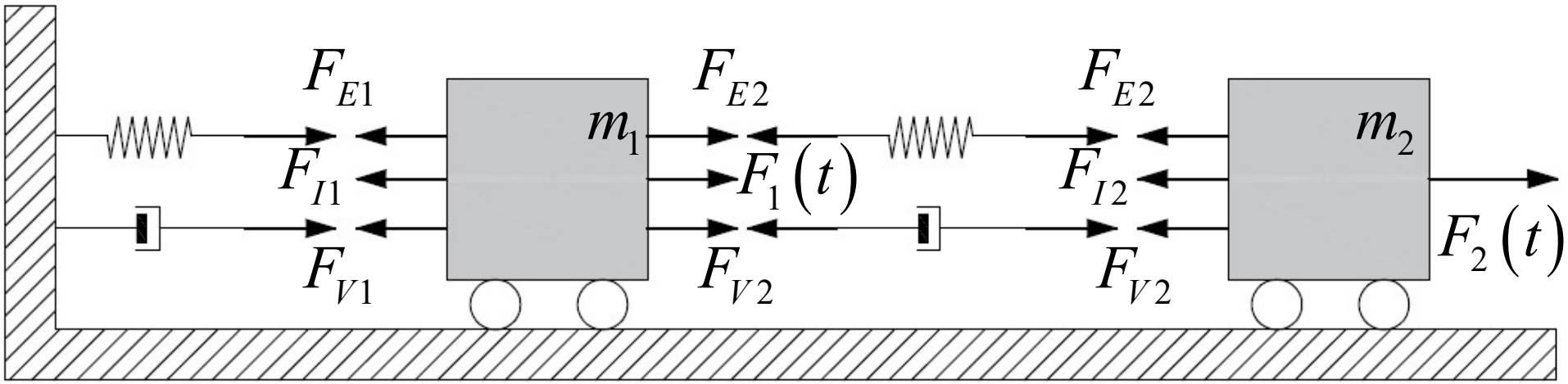


For <u>dynamic equilibrium</u> (d'Alembert's law) one observes the same Figure, which shows all the forces acting on the two masses.

One writes:

$$m_1\ddot{u}_1(t) + c_1\Delta\dot{u}_1(t) + k_1\Delta u_1(t) - c_2\Delta\dot{u}_2(t) - k_2\Delta u_2(t) = F_1(t);$$

$$m_2\ddot{u}_2(t) + c_2\Delta\dot{u}_2(t) + k_2\Delta u_2(t) = F_2(t);$$



<u>Compatibility</u>, now associated to the existence of a "geometry" of the problem, allows one to rewrite the relative displacements Δu_1 andΔu_2 as functions of the two main degrees of freedom $u_1(t)$ and $u_2(t)$:

$$\Delta u_1(t) = \Delta u_1\left(u_1(t), u_2(t)\right) = u_1;$$

$$\Delta u_2(t) = \Delta u_2\left(u_1(t), u_2(t)\right) = u_2 - u_1;$$

so that the final equations of motion are written as follows:

$$m_1\ddot{u}_1(t)+(c_1+c_2)\dot{u}_1(t)-c_2\dot{u}_2(t)+(k_1+k_2)u_1(t)-k_2u_2(t)=F_1(t)$$
$$m_2\ddot{u}_2(t)+c_2\left(\dot{u}_2(t)-\dot{u}_1(t)\right)+k_2\left(u_2(t)-u_1(t)\right)=F_2(t);$$

This is a system of 2 ordinary linear differential equations in the two unknown functions of time $u_1(t)$ and $u_2(t)$.

In order to have a cleaner mathematical expression for this system, it is a good idea to re-write it in matrix form:

$$\mathbf{M}\ddot{\mathbf{u}}(t)+\mathbf{C}\dot{\mathbf{u}}(t)+\mathbf{K}\mathbf{u}(t)=\mathbf{F}(t);$$

$$\mathbf{M}\ddot{\mathbf{u}}(t)+\mathbf{C}\dot{\mathbf{u}}(t)+\mathbf{K}\mathbf{u}(t)=\mathbf{F}(t);$$

In the usual terminolgy for these problems:

$\mathbf{u}(t)=\begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}$ is the vector of the degrees of freedom;

$\mathbf{M}=\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}$ is the mass matrix;

$\mathbf{C}=\begin{bmatrix} c_1+c_2 & -c_2 \\ -c_2 & c_2 \end{bmatrix}$ is the viscous damping matrix;

$\mathbf{K}=\begin{bmatrix} k_1+k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix}$ is the stiffness matrix;

$\mathbf{F}(t)=\begin{bmatrix} F_1(t) \\ F_2(t) \end{bmatrix}$ is the vector of the external forces;

$$\mathbf{M}\ddot{\mathbf{u}}(t)+\mathbf{C}\dot{\mathbf{u}}(t)+\mathbf{K}\mathbf{u}(t)=\mathbf{F}(t);$$

This system of ordinary differential equations must be accompanied by the initial conditions:

$$\mathbf{u}(t=0)=\mathbf{u}_0;$$

$$\dot{\mathbf{u}}(t=0)=\dot{\mathbf{u}}_0;$$

here given by the known, constant vectors $\mathbf{u}_0$ for the initial displacements and $\dot{\mathbf{u}}_0$ for the initial velocities, which we, for simplicity, will always assume to be defined at time t = 0. This is the standard form of the equations of motion of any multi-degree of freedom (MDOF) discrete problem in Dynamics. We must now understand how to write it for a generic discrete system, how to solve it, and which are the main features of its solution.

Before proceeding, however, we must spend few moments for discussing the concept of degrees of freedom, collected into vector $\mathbf{u}(t)$, which are also called **generalized coordinates**, or **lagrangian coordinates**.
In general, in fact, the understanding and choice of the degrees of freedom may pose some difficulties.

A continuous body can be thought of as having an infinite number of degrees of freedom: the motion of all its points (that might be considered as being an infinite number of points) must be described by means of continuous unknown functions (of space and time, in general).
In this case, the unknown displacement function simply describes the displacement vector of each point, and it is an almost intuitive concept.

A discrete body has special features, such as, for example, its being made by rigid parts, possibly connected and/or constrained by simple rigid or elastic components which might introduce a finite number of constraints.
In these cases, the deformed configuration can be fully described by a finite number of functions of time, collected into vector $\mathbf{u}(t)$.

In order to study the motion of such a system in Dynamics, it is first necessary to recognize

- how many are the degrees of freedom,
- and which are the most suitable to be introduced.

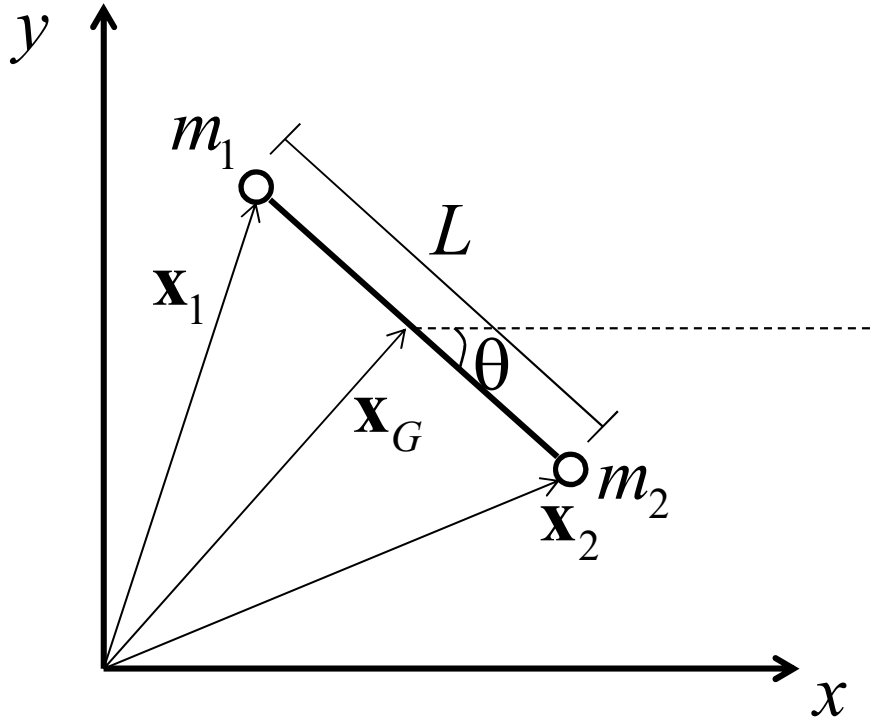
The number of degrees of freedom is uniquely defined by the geometry and constraints of the problem; their choice is in some degree arbitrary.

Number of degrees of freedom

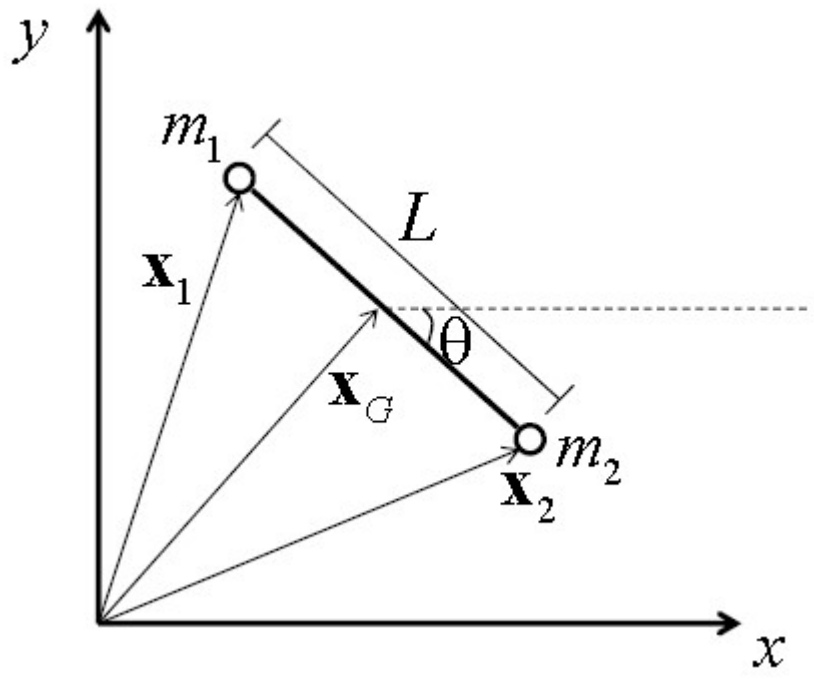
We call degrees of freedom, or generalized coordinates, or lagrangian coordinates of a discrete system a set that includes the minimum number of independent functions of time (scalars, in quasi-static problems) which fully describe the displaced and/or deformed configuration of the system.

In Dynamics, the interest is focused on the motion of all the objects possessing a mass, which can be points, or bars/beams, or plates, or shells, or general 3D solids.
In general, for discrete systems, each such object requires its own set of initial and current coordinates; unfortunately, these coordinates can not be blindly assumed to be the degrees of freedom of the system, because they might be not independent.

Consider in fact, as an example, the following simple system:



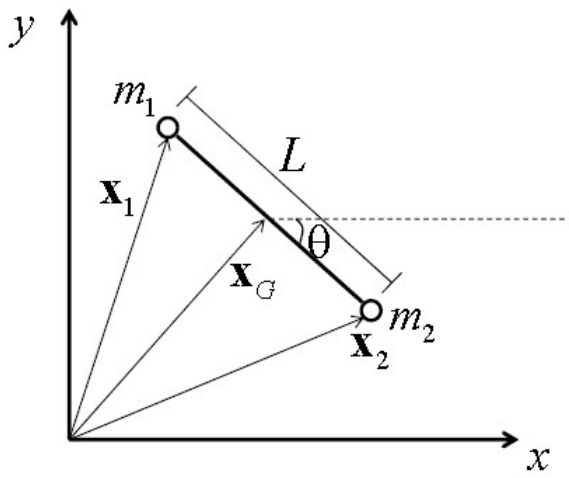
a sort of dumbbell made of two masses *m*1 and *m*2 connected by a massless rigid link of length *L*.



The system is constrained to move in the *x*-*y* plane. Then it seems reasonable to characterize its dynamic motion by means of the two position vectors of the two masses:

$$\mathbf{x}_1(t) = \begin{bmatrix} x_1(t) \\ y_1(t) \end{bmatrix}; \mathbf{x}_2(t) = \begin{bmatrix} x_2(t) \\ y_2(t) \end{bmatrix}$$

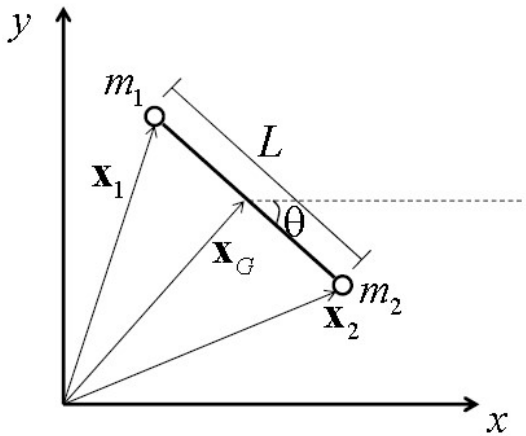
which would seemingly imply the necessity of computing 4 functions of time.



But after a moment one realizes that the 4 coordinates of the two masses are related to each other by the constraint that defines the feature of rigidity of the connecting link:

$$\left(x_2(t)-x_1(t)\right)^2+\left(y_2(t)-y_1(t)\right)^2=L^2$$

a constraint equation that allows one to compute one of the 4 unknowns as a function of the other 3.
Therefore, this problem does not have 4 degrees of freedom, but 3 only, because the knowledge of the values of any 3 coordinates would allow one to compute also the fourth.
This conclusion could be reached in a different way, for this problem: since we have a single rigid body in a plane, with no constraints, its motion must be described by 3 degrees of freedom only.



$$x_1 = x_G - \frac{L}{2} cos\,\theta; \quad y_1 = y_G + \frac{L}{2} sin\,\theta;$$

$$x_2 = x_G + \frac{L}{2} cos\,\theta; \quad y_2 = y_G - \frac{L}{2} sin\,\theta;$$

For example, in this case one could choose, as degrees of freedom, the position of the center of mass *G*, defined by vector $\mathbf{x}_G(t)$, with 2 scalar components, and the rotation of the link, defined by the angle $\theta(t)$, a third scalar.

In most situations, it is much more convenient to identify and choose properly a correct set of degrees of freedom, rather than introducing a larger number of non fully independent coordinates and understanding the relevant constraint equations. For our example, besides, the constraint equation between x_1, x_2 , y_1, and y_2 was nonlinear, which would add a fully unnecessary complication.

As we have seen by means of the simple initial example, the equations of motion for a MDOF system with *N* degrees of freedom can always be written, in matrix form, as follows:

$$\mathbf{M}\ddot{\mathbf{u}}(t)+\mathbf{C}\dot{\mathbf{u}}(t)+\mathbf{K}\mathbf{u}(t)=\mathbf{F}(t);$$

$$\mathbf{u}(0)=\mathbf{u}_0;\dot{\mathbf{u}}(0)=\dot{\mathbf{u}}_0;$$

u(*t*) is the *N*x1 vector that collects the *N* degrees of freedom;
M is the mass matrix, *N*x*N*;
C is the viscous damping matrix, *N*x*N*;
K is the stiffness matrix, *N*x*N*;
and **F**(*t*) is the *N*x1 vector of the external forces.
This is a system of linear, second-order ordinary differential equations in the unknown vector **u**(*t*), whose components are all functions of time only. The initial values of **u**(*t*) are contained into vectors **u**0 and **v**0, having constant known values.

The mass matrix **M** is symmetric and positive definite.

The stiffness matrix **K** is symmetric and, if the system if suitably constrained to the reference system, it is also positive definite.
The stiffness matrix **K** is in fact the same matrix that governs the equilibrium of the same discrete body under quasi-static conditions in the presence of time independent loading and without viscous damping:

$$\mathbf{Ku} = \mathbf{F}; \Rightarrow \mathbf{u} = \mathbf{K}^{-1}\mathbf{F};$$

a system of linear algebraic equations that can be solved if **K** is not singular, i.e., if the system can not undergo any rigid-body motion.

In Dynamics, however, rigid body motions are just a perfectly possible special case of the most general type of motion.

The body under investigation can well have no constraints at all to the reference system, and its stiffness matrix can be singular.

In Dynamics, even if the stiffness matrix is singular "equilibrium" is anyway reached under dynamic conditions, i.e., in the presence of inertia forces associated to some type of motion.

This situation, which would make impossible the solution in a quasi-static case, can be fully accounted for in Dynamics.

1. Free vibrations (no applied force) of an undamped MDOF system.

We start by studying the solution for the special case of a free motion, determined by the initial conditions only, in the undamped case. Then the equations of motion reduce to:

$$\mathbf{M}\ddot{\mathbf{u}}(t)+\mathbf{K}\mathbf{u}(t)=\mathbf{0};$$

$$\mathbf{u}(0)=\mathbf{u}_0;\dot{\mathbf{u}}(0)=\dot{\mathbf{u}}_0;$$

In all the following, we assume to be able to compute and assemble the elements of all the matrices, and, if it exists, of the external force vector, that govern the equations of motion. In order to compute a solution of these homogeneous equations of motion, we assume that this solution would describe a harmonic motion, by analogy with the behavior of SDOF systems under initial conditions only.

Next, we assume, as a tentative solution, a special type of harmonic motion – the simplest possible, in fact – in which all the components of vector **u**(t) have the same time dependence, but each component has its own amplitude.

This special tentative solution of the harmonic type, which will be denoted by **z**(t), can therefore be expressed as follows:

$$\mathbf{z}(t) = \boldsymbol{\psi}\, sin(\omega t + \theta)$$

In this tentative solution, $\boldsymbol{\psi}$ is the vector defining the shape of the system during the motion, which is time independent since all the components of **z**(t) are assumed to change in the same way with time; ω is a frequency and θ is a phase angle.

This special tentative solution is explored in order to check
(a) if it is possible to satisfy with this simple choice the equations of motion, and
(b) if yes, under what conditions.

This tentative solution defines a harmonic motion in which the discrete system is displaced with a specific constant shape defined by the constant vector $\boldsymbol{\psi}$; this shape changes its amplitude with time according to a sine function with frequency ω and phase angle θ. In order to check if the solution defined by $\mathbf{z}(t)$ satisfies the equations of motion, we replace it into them, obtaining what follows:

$$-\omega^2 \mathbf{M}\boldsymbol{\psi}\, sin\left(\omega t + \theta\right) + \mathbf{K}\boldsymbol{\psi}\, sin\left(\omega t + \theta\right) = \mathbf{0}$$

Since the sine function takes arbitrary values, it can be simplified from this result, which yields:

$$\left[\mathbf{K}-\omega^2\mathbf{M}\right]\boldsymbol{\psi}=\mathbf{0}$$

a very well known mathematical problem in 2 unknowns: the shape vector Ψ and the frequency ω.

This is an **eigenvalue problem**.

This is in fact a system of linear algebraic equations in the unknown vector Ψ with zero known term, i.e., a homogeneous linear system. It always has the trivial solution $\boldsymbol{\psi}=\mathbf{0}$ but it also admits non-zero solutions if:

$$det\left[\mathbf{K}-\omega^2\mathbf{M}\right]=0$$

This equation is called the **frequency equation** of the MDOF system with *N* degrees of freedom.

Expanding the above determinant one obtains a scalar algebraic equation of the *N*-th degree in the frequency parameter ω^2, which has *N* roots, all real and non-negative in "simple" engineering problems because the governing matrices are both symmetric and positive semidefinite.

These *N* roots, i.e., the *N* values of the frequency ω^2 that satisfy the frequency equation, are the **eigenvalues** of the problem.

Replacing each eigenvalue into the starting homogeneous linear system, namely

$$\left[\mathbf{K} - \omega^2 \mathbf{M}\right]\boldsymbol{\psi} = \mathbf{0}$$

solving the system *N* times, one finds *N* vectors $\boldsymbol{\psi}$, called the **eigenvectors** of the problem. Each of them is defined up to a constant: it has a precise shape, but arbitrary magnitude.

The *N* eigenvalues, i.e., the values of the parameter ω^2 that satisfy the frequency equation, are denoted by

$$\omega_1^2, \omega_2^2, \omega_3^2, \ldots, \omega^2{}_N$$

and their *N* square roots are called the **system natural frequencies**

$$\omega_i, i = 1, \ldots, N$$

The system natural frequencies ω_i are normally all distinct from one another, once more because of the features of the mass and stiffness matrices. In some special cases, called "degenerate", however, some frequencies may be coincident, with $\omega_i = \omega_{i+1}$, or even with higher multiplicity.

To each natural frequency a corresponding time period can be associated through the usual relationship:

$$T_i = \frac{2\pi}{\omega_i}, i = 1, \ldots, N$$

These are the **natural time periods** of the system.

The natural frequencies are usually arranged in increasing order of numerical value:

$$\omega_1 \leq \omega_2 \leq \omega_3 \leq \ldots \leq \omega_N$$

The lowest frequency ω_1 is called **fundamental frequency** of the system. For many practical problems, it is the most important natural frequency.

Since each natural frequency ω_i corresponds to a single eigenvector $\Psi_{(i)}$, the *N* eigenvectors $\Psi_{(i)}$ are called **natural modes** or **modal vectors** of the MDOF system.

Recall that the shape of each natural mode is unique, but its amplitude is undefined.
In order to define the complete motion $\mathbf{z}_{(i)}(t)$ associated to each of these N special solutions, we need to somehow set a unique value for the amplitudes of the natural modes.

The process of setting an amplitude for eigenvectors is called **normalization**. Any eigenvector can be written in the following mathematical form:

$$\Psi_{(i)} = \alpha_i \hat{\Psi}_{(i)}$$

Where $\hat{\psi}_{(i)}$ is a shape vector (an eigenvector), and α_i is its undefined amplitude, which we want to define using some normalization technique.

Therefore, the normalization of each eigenvector implies solving a problem in a single unknown: the amplitude α_i.

- The easiest and most trivial normalization technique consists in setting equal to 1 the largest component of each eigenvector, and redefining all the other components correspondingly.

- Another trivial normalization technique prescribes that each eigenvector has modulus equal to 1, i.e., computes its amplitude from the scalar quadratic equation

$$\psi_{(i)}{}^T \psi_{(i)} = 1, \quad \text{i=1,...,N}$$

$$\psi_{(i)} = \alpha_i \hat{\psi}_{(i)} \Rightarrow \alpha_i^2 \hat{\psi}_{(i)}{}^T \hat{\psi}_{(i)} = 1 \Rightarrow \alpha_i = \frac{1}{\sqrt{\hat{\psi}_{(i)}{}^T \hat{\psi}_{(i)}}} = \frac{1}{\left|\hat{\psi}_{(i)}\right|}$$

- A slightly more complex, but convenient normalization technique consists in computing the amplitude of each eigenvector from the following scalar quadratic equation:

$$\boldsymbol{\psi}_{(i)}{}^{T}\mathbf{M}\boldsymbol{\psi}_{(i)} = 1, \quad \mathrm{i=1,...,N}$$

$$\boldsymbol{\psi}_{(i)} = \alpha_i \hat{\boldsymbol{\psi}}_{(i)} \Rightarrow \alpha_i^2 \hat{\boldsymbol{\psi}}_{(i)}{}^{T}\mathbf{M}\hat{\boldsymbol{\psi}}_{(i)} = 1 \Rightarrow \alpha_i = \frac{1}{\sqrt{\hat{\boldsymbol{\psi}}_{(i)}{}^{T}\mathbf{M}\hat{\boldsymbol{\psi}}_{(i)}}}$$

in which the mass matrix **M** has been used as a "normalization coefficient". In this case eigenvectors are said to be **mass-normalized**.

By adopting this form of normalization one can also prove what follows:

$$\boldsymbol{\psi}_{(i)}{}^{T}\mathbf{K}\boldsymbol{\psi}_{(i)} = \omega^2{}_i, \quad \mathrm{i=1,...,N}$$

which descends immediately from the starting system of equations:

$$\mathbf{K}\psi_{(i)} - \omega^2{}_i\mathbf{M}\psi_{(i)} = \mathbf{0}, \quad \mathrm{i=1,...,N}$$

$$\psi_{(i)}{}^T\mathbf{K}\psi_{(i)} - \omega^2{}_i\psi_{(i)}{}^T\mathbf{M}\psi_{(i)} = 0, \quad \mathrm{i=1,...,N}$$

$$\psi_{(i)}{}^T\mathbf{K}\psi_{(i)} - \omega^2{}_i = 0, \quad \mathrm{i=1,...,N}$$

This form of normalization is alsoconvenient in view of a very importantfeature of the eigenvectors $\psi_{(i)}$: their **orthogonality** with respect to both the mass matrix and the stiffness matrix.

Mathematically, orthogonality of two vectors $\Psi_{(i)}$ and $\Psi_{(j)}$ with respect to both the mass matrix and the stiffness matrix means what follows:

$$\psi_{(i)}{}^T\mathbf{M}\psi_{(j)} = 0, \quad \mathrm{i} \neq \mathrm{j}, \omega_i \neq \omega_j$$

$$\psi_{(i)}{}^T\mathbf{K}\psi_{(j)} = 0, \quad \mathrm{i} \neq \mathrm{j}, \omega_i \neq \omega_j$$

This property can be proved as follows. Consider two distinct solutions of the eigenvalue problem ω_i and ω_j and $\psi_{(i)}$ and $\psi_{(j)}$ with non-coinciding eigenvalues, i.e.,$\omega_i \neq \omega_j$.

Both these two solutions satisfy the starting system of equations

$$\mathbf{K}\psi_{(i)} = \omega^2{}_i\mathbf{M}\psi_{(i)}; \quad \mathbf{K}\psi_{(j)} = \omega^2{}_j\mathbf{M}\psi_{(j)};$$

Premultiply the first by $\psi_{(j)}{}^T$ and the second by $\psi_{(i)}{}^T$:

$$\psi_{(j)}{}^T\mathbf{K}\psi_{(i)} = \omega^2{}_i\psi_{(j)}{}^T\mathbf{M}\psi_{(i)}; \quad \psi_{(i)}{}^T\mathbf{K}\psi_{(j)} = \omega^2{}_j\psi_{(i)}{}^T\mathbf{M}\psi_{(j)};$$

Transpose the second one:

$$\left(\psi_{(i)}{}^T\mathbf{K}\psi_{(j)}\right)^T = \psi_{(j)}{}^T\mathbf{K}\psi_{(i)} = \omega^2{}_j\left(\psi_{(i)}{}^T\mathbf{M}\psi_{(j)}\right)^T = \omega^2{}_j\psi_{(j)}{}^T\mathbf{M}\psi_{(i)};$$

Subtracting this last equation from the first one gets:

$$0 = \left(\omega^2{}_i - \omega_j^2\right)\boldsymbol{\psi}_{(j)}{}^T\mathbf{M}\boldsymbol{\psi}_{(i)} = \left(\omega^2{}_i - \omega_j^2\right)\boldsymbol{\psi}_{(i)}{}^T\mathbf{M}\boldsymbol{\psi}_{(j)}$$

which, for $\omega_i \neq \omega_j$, proves the first orthogonality relationship.

To prove the orthogonality also with respect to the stiffness matrix, one simply recalls that:

$$\left(\boldsymbol{\psi}_{(i)}{}^T\mathbf{K}\boldsymbol{\psi}_{(j)}\right)^T = \boldsymbol{\psi}_{(j)}{}^T\mathbf{K}\boldsymbol{\psi}_{(i)} = \omega^2{}_j\boldsymbol{\psi}_{(j)}{}^T\mathbf{M}\boldsymbol{\psi}_{(i)} = 0;$$

$$\Rightarrow \boldsymbol{\psi}_{(i)}{}^T\mathbf{K}\boldsymbol{\psi}_{(j)} = 0$$

If the natural modes have been also normalized, they are called **orthonormal**.

Finally, if they have been normalized with respect to the mass matrix, then they satisfy the following nice relationships,

$$\boldsymbol{\psi}_{(i)}{}^{T}\mathbf{M}\boldsymbol{\psi}_{(j)} = \delta_{ij}$$

$$\boldsymbol{\psi}_{(i)}{}^{T}\mathbf{K}\boldsymbol{\psi}_{(j)} = \omega^2{}_i\delta_{ij} = \omega^2{}_j\delta_{ij}$$

where δ_{ij} is the Kronecker Delta.

We have thus discovered not just one, but a set of *N* independent special solutions $\mathbf{z}_{(i)}(t)$ of the equations of motion:

$$\mathbf{z}_{(i)}(t) = \boldsymbol{\psi}_{(i)}\, sin\left(\omega_i t + \theta_i\right), \quad i = 1,\ldots,N$$

each with its own shape $\Psi_{(i)}$, frequency ω_i ,and phase angle θ_i (the phase angles are all undetermined yet, though). They describe the free vibrations of the system according to its natural modes. These *N* solutions $\mathbf{z}_{(i)}(t)$ are called **natural solutions**.

The shape vectors $\Psi_{(i)}$ and the frequencies ω_i of these *N* independent special solutions are not related to any initial condition: they depend only on the material and geometry properties of the system, i.e., from its mass and stiffness matrices. The phase angles θ_i are left undetermined in this representation and we will see that they depend upon initial conditions and frequencies ω_i.

We may now ask if the knowledge of the N vectors $\mathbf{z}_{(i)}(t)$, and in particular of their shapes $\psi_{(i)}$ and the associated frequencies ω_i, can be useful to compute the actual response of the system to any set of prescribed initial conditions.

The answer is YES: the superposition principle,valid for any linear problem such as the oneunder study, allows us to express a generalsolution $\mathbf{u}(t)$, for the free motion of an undamped MDOF system, as a linearcombination of its natural solutions $\mathbf{z}_{(i)}(t)$:

$$\mathbf{u}(t) = \sum_{i=1}^{N} C_i \mathbf{z}_{(i)}(t) = \sum_{i=1}^{N} C_i \psi_{(i)} \, sin(\omega_i t + \theta_i)$$

in which both the coefficients C_i and the phase angles θ_i must be computed from the initial conditions.

Note that, as is obvious, the values of both C_i and θ_i depend also on the normalization technique adopted for the eigenvectors $\psi_{(i)}$.

We now proceed to computing in full this complete general solution. To this purpose, it is convenient to re-write the solution $\mathbf{u}(t)$ as a sum of sine and cosine terms:

$$\mathbf{u}(t) = \psi_{(i)}\left(A_i \sin \omega_i t + B_i \cos \omega_i t\right)$$

where

$$\sin\left(\omega_i t + \theta_i\right) = \sin \omega_i t \cos \theta_i + \cos \omega_i t \sin \theta_i$$

$$\Rightarrow A_i = C_i \cos \theta_i ; B_i = C_i \sin \theta_i ;$$

Prescribing the initial conditions one obtains:

$$\mathbf{u}(0) = \sum_{i=1}^{N} \psi_{(i)} B_i = \mathbf{u}_0 ; \quad \dot{\mathbf{u}}(0) = \sum_{i=1}^{N} \psi_{(i)} \omega_i A_i = \dot{\mathbf{u}}_0 ;$$

If vectors $\psi_{(i)}$ were generic vectors, the calculation of the values for the N constants A_i and B_i would require the solution of two linear systems; their expression in a simple analytical form would therefore be impossible.

But if vectors $\psi_{(i)}$ are orthonormal eigenvectors, we can exploit their properties as follows.

Premultiply both initial conditions by $\psi_{(j)}{}^T\mathbf{M}$:

$$\psi_{(j)}{}^T\mathbf{M}\sum_{i=1}^{N}\psi_{(i)}B_i = \psi_{(j)}{}^T\mathbf{M}\mathbf{u}_0; \quad \psi_{(j)}{}^T\mathbf{M}\sum_{i=1}^{N}\psi_{(i)}\omega_i A_i = \psi_{(j)}{}^T\mathbf{M}\dot{\mathbf{u}}_0;$$

Now the orthogonality of the eigenvectors with respect to the mass matrix, and their normalization with respect to the same matrix, leads to the following:

$$\psi_{(i)}{}^T\mathbf{M}\psi_{(j)} = \delta_{ij}; \psi_{(j)}{}^T\mathbf{M}\sum_{i=1}^{N}\psi_{(i)}B_i = B_j; \psi_{(j)}{}^T\mathbf{M}\sum_{i=1}^{N}\psi_{(i)}\omega_i A_i = \omega_j A_j;$$

Therefore, we can finally write:

$$B_i = \boldsymbol{\psi}_{(i)}{}^T \mathbf{M}\mathbf{u}_0; \quad A_i = \frac{\boldsymbol{\psi}_{(i)}{}^T \mathbf{M}\dot{\mathbf{u}}_0}{\omega_i};$$

and the complete solution for the free motion of an undamped MDOF system, for any given set of initial conditions, reads then as follows:

$$\mathbf{u}(t) = \sum_{i=1}^{N} \boldsymbol{\psi}_{(i)} \left(\frac{\boldsymbol{\psi}_{(i)}{}^T \mathbf{M}\dot{\mathbf{u}}_0}{\omega_i} sin\, \omega_i t + \boldsymbol{\psi}_{(i)}{}^T \mathbf{M}\mathbf{u}_0\, cos\, \omega_i t \right);$$

$\boldsymbol{\psi}_{(i)}$ being mass-normalized eigenvectors.

Total energy associated to the free motion of an undamped MDOF system

As was done for the SDOF case, also for the MDOF one it is possible to compute the energy associated to the forces and displacements involved by the motion. One way to do this is to pass through the definition of the power, writing what follows:

$$\mathcal{P}(t) = \dot{\mathbf{u}}^T(t)\left[\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{K}\mathbf{u}(t)\right]$$

The energy is obtained integrating the power with respect to time. Computing only the primitive function, and recalling that both the mass and the stiffness matrices are symmetric, one has:

$$E(t) = \int_t \mathcal{P}(t)dt = \int_t \dot{\mathbf{u}}^T(t)\mathbf{M}\ddot{\mathbf{u}}(t)\,dt + \int_t \dot{\mathbf{u}}^T(t)\mathbf{K}\mathbf{u}(t)\,dt =$$

$$= \int_{\dot{\mathbf{u}}(t)} \dot{\mathbf{u}}^T(t)\mathbf{M}d\dot{\mathbf{u}} + \int_t \dot{\mathbf{u}}^T(t)\mathbf{K}\mathbf{u}(t)\,dt =$$

$$= \frac{1}{2}\dot{\mathbf{u}}^T(t)\mathbf{M}\dot{\mathbf{u}}(t) + \int_{\mathbf{u}(t)} \mathbf{u}^T(t)\mathbf{K}d\mathbf{u} =$$

$$= \frac{1}{2}\dot{\mathbf{u}}^T(t)\mathbf{M}\dot{\mathbf{u}}(t) + \frac{1}{2}\mathbf{u}^T(t)\mathbf{K}\mathbf{u}(t) = \mathcal{T}(t) + \mathcal{V}(t)$$

As expected, one obtains a result completely analogous to the one derived for the SDOF case: the total energy is the sum of a kinetic and a strain energy.

$$\text{Total energy:} \quad \mathcal{T}(t) + \mathcal{V}(t) = \frac{1}{2}\dot{\mathbf{u}}^T(t)\mathbf{M}\dot{\mathbf{u}}(t) + \frac{1}{2}\mathbf{u}^T(t)\mathbf{K}\mathbf{u}(t)$$

The same conservation theorem holds as well.

2. Undamped forced motion for a MDOF system

The equations of motion for an undamped MDOF system, in the presence of an external excitation defined by a known vector **F**(*t*), are:

$$\mathbf{M}\ddot{\mathbf{u}}(t)+\mathbf{K}\mathbf{u}(t)=\mathbf{F}(t);$$
$$\mathbf{u}(0)=\mathbf{u}_0;\dot{\mathbf{u}}(0)=\dot{\mathbf{u}}_0;$$

The presence of the forcing term **F**(*t*) makes it impossible to obtain a general solution using the superposition technique developed for the free motion.
In general, since both the mass matrix **M** and the stiffness matrix **K** are not diagonal, the equations of motion are coupled, and their analytical solution is by no means an easy task.

In order to simplify the problem, at least in principle, one can try to understand if and how the equations of motion can be made uncoupled.
If this operation is possible, in fact, each new independent equation reduces to one of the SDOF type, and it can be solved through the techniques illustrated in the previous part of this class.
This uncoupling operation corresponds to transforming the original equations

$$\mathbf{M}\ddot{\mathbf{u}}(t)+\mathbf{K}\mathbf{u}(t)=\mathbf{F}(t);$$

into a set of new but equivalent ones

$$\tilde{\mathbf{M}}\ddot{\mathbf{p}}(t)+\tilde{\mathbf{K}}\mathbf{p}(t)=\tilde{\mathbf{F}}(t);$$

in which $\mathbf{p}(t)$ is a new set of degrees of freedom, and matrices $\widetilde{\boldsymbol{M}}$ and $\widetilde{\boldsymbol{K}}$ are diagonal.

The new set of degrees of freedom, **p**(*t*), is conveniently obtained from a linear transformation operated on the original set **u**(*t*) as follows:

$$\mathbf{u}(t) = \mathbf{T}\mathbf{p}(t);$$
$$\mathbf{p}(t) = \mathbf{T}^{-1}\mathbf{u}(t);$$

where the symbol **T** denotes a constant, non-singular square matrix called transformation matrix.
Having suitably defined the transformation matrix **T** one can immediately write the transformed form of the equations of motion:

$$\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{K}\mathbf{u}(t) = \mathbf{F}(t);$$
$$\mathbf{u}(t) = \mathbf{T}\mathbf{p}(t); \dot{\mathbf{u}}(t) = \mathbf{T}\dot{\mathbf{p}}(t); \ddot{\mathbf{u}}(t) = \mathbf{T}\ddot{\mathbf{p}}(t);$$

$$\mathbf{T}^T\mathbf{M}\mathbf{T}\ddot{\mathbf{p}}(t)+\mathbf{T}^T\mathbf{K}\mathbf{T}\mathbf{p}(t)=\mathbf{T}^T\mathbf{F}(t);$$

$$\tilde{\mathbf{M}}\ddot{\mathbf{p}}(t)+\tilde{\mathbf{K}}\mathbf{p}(t)=\tilde{\mathbf{F}}(t);$$

$$\tilde{\mathbf{M}}=\mathbf{T}^T\mathbf{M}\mathbf{T};\quad \tilde{\mathbf{K}}=\mathbf{T}^T\mathbf{K}\mathbf{T};\quad \tilde{\mathbf{F}}(t)=\mathbf{T}^T\mathbf{F}(t);$$

Given a transformation matrix **T**, the corresponding transformed mass and stiffness matrices could have been obtained also from the definition of the kinetic and strain energies:

$$\mathcal{T}(t)=\frac{1}{2}\dot{\mathbf{p}}^T(t)\left[\mathbf{T}^T\mathbf{M}\mathbf{T}\right]\dot{\mathbf{p}}(t)=\frac{1}{2}\dot{\mathbf{p}}^T(t)\tilde{\mathbf{M}}\dot{\mathbf{p}}(t)$$

$$\mathcal{V}(t)=\frac{1}{2}\mathbf{p}^T(t)\left[\mathbf{T}^T\mathbf{K}\mathbf{T}\right]\mathbf{p}(t)=\frac{1}{2}\mathbf{p}^T(t)\tilde{\mathbf{K}}\mathbf{p}(t)$$

We now ask ourselves if it exists a special transformation matrix **T** that, once applied to define transformed mass and stiffness matrices, makes them diagonal, so that the transformed equations of motion are uncoupled.
If this transformation exists, it means that it exists a special set of degrees of freedom **p**(*t*) whose components can be computed one at a time, independently from each other.

It turns out that such a special transformation matrix **T** does indeed exist. It can be obtained after having introduced the properties of orthonormality of the modal vectors, or natural modes, $\boldsymbol{\Psi}_{(i)}$, of the system. Recall in fact that, using the mass-based normalization, the following properties hold:

$$\boldsymbol{\psi}_{(i)}{}^{T}\mathbf{M}\boldsymbol{\psi}_{(j)} = \delta_{ij};$$

$$\boldsymbol{\psi}_{(i)}{}^{T}\mathbf{K}\boldsymbol{\psi}_{(j)} = \delta_{ij}\omega_i^2 = \delta_{ij}\omega_j^2;$$

Then, it is possible to prove that a transformation matrix $\mathbf{T}_M$ made by all the natural mode vectors does indeed produce the uncoupling of the equations of motion

$$\mathbf{T}_M = \begin{bmatrix} \boldsymbol{\psi}_{(1)} & \boldsymbol{\psi}_{(2)} & \cdots & \boldsymbol{\psi}_{(N)} \end{bmatrix}$$

This matrix $\mathbf{T}_M$ is called **modal matrix**.
The procedure for solving the equations of motion by transforming them into a system of uncoupled equations through the modal matrix is called **modal analysis**.

Let us see what is the aspect of the transformed mass and stiffness matrices if the transformation matrix is the modal matrix $\mathbf{T}_M$ made by all the natural modes.

Restarting from the eigenvalue problem that governs the free motion of the same MDOF system, we know that, for each eigenvalue ω_i^2 and eigenvector $\boldsymbol{\psi}_{(i)}$ couple, the following identity holds:

$$\mathbf{K}\boldsymbol{\psi}_{(i)} = \omega_i^2 \mathbf{M}\boldsymbol{\psi}_{(i)}$$

which is an identity between two vectors.
Since it holds for all i = 1, …, N, we can put all these vectors together at both sides and write the following matrix identity:

$$\mathbf{K}\mathbf{T}_M = \mathbf{M}\mathbf{T}_M \boldsymbol{\Omega}$$

where $\boldsymbol{\Omega}$ is the following diagonal matrix:

$$\boldsymbol{\Omega} = \begin{bmatrix} \omega_1^2 & 0 & \dots & 0 \\ 0 & \omega_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \omega_N^2 \end{bmatrix}$$

Similarly, putting together all the orthogonality properties of the natural modes, one can write:

$$\boldsymbol{\psi}_{(i)}{}^{T}\mathbf{M}\boldsymbol{\psi}_{(j)} = \delta_{ij}; \Rightarrow \mathbf{T}_M^T\mathbf{M}\mathbf{T}_M = \mathbf{I};$$

$$\boldsymbol{\psi}_{(i)}{}^{T}\mathbf{K}\boldsymbol{\psi}_{(j)} = \delta_{ij}\omega_i^2 = \delta_{ij}\omega_j^2; \Rightarrow \mathbf{T}_M^T\mathbf{K}\mathbf{T}_M = \boldsymbol{\Omega};$$

Consider now the transformed matrices that govern the transformed equations of motion in the new set of degrees of freedom **p**(*t*).

$$\tilde{\mathbf{M}} = \mathbf{T}_M^T\mathbf{M}\mathbf{T}_M = \mathbf{I};$$

$$\tilde{\mathbf{K}} = \mathbf{T}_M^T\mathbf{K}\mathbf{T}_M = \boldsymbol{\Omega};$$

Then the transformed form of the equations of motion, in this new set of degrees of freedom **p**(*t*), is the following uncoupled one:

$$\ddot{\mathbf{p}}(t) + \boldsymbol{\Omega}\mathbf{p}(t) = \mathbf{T}_M^T\mathbf{F}(t) = \tilde{\mathbf{F}}(t);$$

$$\ddot{\mathbf{p}}(t)+\mathbf{\Omega}\mathbf{p}(t)=\mathbf{T}_M^T\mathbf{F}(t)=\tilde{\mathbf{F}}(t);$$

$$\ddot{p}_i(t)+\omega_i^2 p_i(t)=\tilde{F}_i(t) \quad \text{i=1,...,N}$$

This is a set of *N* independent modal equations, each of which can be "easily" solved, one at a time, for each component $p_i(t)$, adopting the techniques studied for the SDOF systems. Once vector $\mathbf{p}(t)$ is computed, the original set of degrees of freedom $\mathbf{u}(t)$ can be obtained from the coordinate transformation

$$\mathbf{u}(t)=\mathbf{T}_M^T\mathbf{p}(t);$$

Since we already know the features of the solution for the SDOF systems, we will not go in detail into the same results for uncoupled MDOF systems.

Here we want to remind only an important feature of the motion of MDOF systems under a harmonic external excitation: any time the external excitation has a frequency η coinciding with one of the natural frequencies ω_i of the MDOF system, resonance occurs.

3 Damped forced motion for a MDOF system

The general equations of motion for a viscouslydamped, linear elastic MDOF system, in the presence of an external excitation defined by vector **F**(*t*), are:

$$\mathbf{M}\ddot{\mathbf{u}}(t)+\mathbf{C}\dot{\mathbf{u}}(t)+\mathbf{K}\mathbf{u}(t)=\mathbf{F}(t);$$

$$\mathbf{u}(0)=\mathbf{u}_0;$$

$$\dot{\mathbf{u}}(0)=\dot{\mathbf{u}}_0;$$

In this equation, **C** represents the viscous damping matrix. We might hope to recover the analysis developed for the undamped case, i.e., to transform the degrees of freedom **u**(*t*) to the set of the natural ones **p**(*t*), by means of the modal matrix **T***M*, in order to obtain an uncoupled system of differential equations.

This procedure would immediately lead to the following results:

$$\mathbf{u}(t) = \mathbf{T}_M \mathbf{p}(t);$$

$$\mathbf{T}_M = \begin{bmatrix} \boldsymbol{\psi}_{(1)} & \boldsymbol{\psi}_{(2)} & \cdots & \boldsymbol{\psi}_{(N)} \end{bmatrix};$$

$$\ddot{\mathbf{p}}(t) + \mathbf{T}_M{}^T \mathbf{C} \mathbf{T}_M \dot{\mathbf{p}}(t) + \boldsymbol{\Omega}\mathbf{p}(t) = \mathbf{T}_M{}^T \mathbf{F}(t);$$

$$\ddot{\mathbf{p}}(t) + \tilde{\mathbf{C}}\dot{\mathbf{p}}(t) + \boldsymbol{\Omega}\mathbf{p}(t) = \tilde{\mathbf{F}}(t);$$

where $\boldsymbol{\psi}_{(i)}$ are the same eigenvectors of the undamped system, i.e., the natural modes.

Nothing can be said, in general, about the features of the transformed damping matrix even in the case of a modal transformation. Therefore, in the presence of a generic damping matrix there is no guarantee, even expressing the motion in its natural coordinates, that the governing equations become uncoupled.

If, however, a special form of the damping matrix **C** makes the transformed one diagonal, then everything is again reduced to the analysis of *N* uncoupled equations of motion. For this to happen, the following orthogonality condition also with respect to **C** should hold:

$$\boldsymbol{\psi}_{(i)}{}^{T}\mathbf{C}\boldsymbol{\psi}_{(j)} = 0 \quad \forall i \neq j$$

The practical determination of suitable forms for the damping matrix **C**, and the evaluation of realistic numerical values for its components, are both in any event no trivial tasks. Therefore, in practice (for example in Finite Element analyses) a form for the damping matrix **C** which satisfies this “artificial” orthogonality condition is often adopted.

One of the most commonly adopted such forms was proposed by Lord Rayleigh (John William Strutt) in 1877, in which the damping matrix **C** is defined as follows:

$$\mathbf{C}_R = \alpha_R \mathbf{M} + \beta_R \mathbf{K}$$

where α_R and β_R are two constant coefficients, usually with very small values, that are used to linearly combine the mass and stiffness matrices in order to produce a damping matrix. This special form of damping is called **proportional damping**, or **Rayleigh damping**.

For each natural mode, it is easy to verify that the following relationship holds between the Rayleigh damping parameters α_R and β_R, the natural frequency ω_i, and the viscous damping factor $0 \leq \zeta_i \leq 1$:

$$\zeta_i = \frac{\alpha_R}{2\omega_i} + \frac{\beta_R \omega_i}{2}$$

In this way, the transformed equations of motion, always using the modal matrix $\mathbf{T}_M$ to transform the degrees of freedom, become once more uncoupled even in the presence of viscous damping:

$$\mathbf{C}_R = \alpha_R \mathbf{M} + \beta_R \mathbf{K}$$

$$\Rightarrow \ddot{\mathbf{p}}(t) + \left[\alpha_R \mathbf{I} + \beta_R \mathbf{\Omega}\right]\dot{\mathbf{p}}(t) + \mathbf{\Omega}\mathbf{p}(t) = \tilde{\mathbf{F}}(t);$$

From a practical viewpoint, the analysis of MDOF systems requires ability in writing down the governing matrices **M**, **C**, and **K**; after this, it's a matter of Mathematics.

Of course, in the presence of a large number of degrees of freedom the calculation of a complete analytical solution becomes very impractical.

In past times, several approximate techniques to obtain reasonable solutions analytically have been developed; among them, the Stodola method and the Rayleigh method.

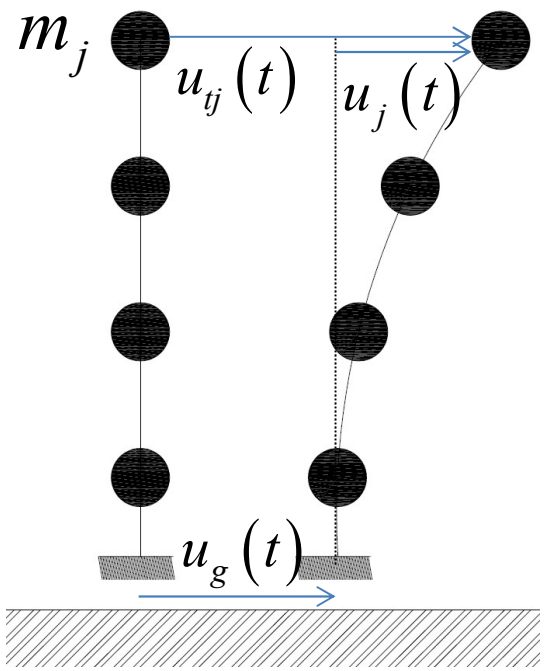
In current times, the use of numerical methods is common practice. If one can compute numerically eigenvalues and eigenvectors, one has all that he needs to determine solutions for any MDOF system.

In many cases, moreover, a reasonably good solution at a low cost can be obtained by superposing only a small number of natural modes, starting from the one associated to the lowest natural frequency ω_1 and moving on for increasing frequency number. This is because in normal Engineering structures the <u>motion is dominated by the low-frequency modes</u>.

In many other cases – always when the motion is non-linear, but often also for a linear motion, when the eigenvalue/eigenvector superposition is not convenient, or in other special situations, such as for instance those of wave propagation, or of “fast” dynamics problems – it has become common practice to adopt techniques for the direct integration in time of the governing equations for MDOF systems.

We will now briefly summarize how a modal analysis is performed, for MDOF systems, for the special case of an input defined only as a prescribed ground motion, i.e., the input typical in a seismic Structural Dynamics analysis.
In the following, for simplicity, we assume that the ground motion is the same at all the structure points connected to the ground.

Consider a MDOF system dynamically loaded by a given motion $u_g(t)$ of its support, defined with reference to a fixed reference system.



$$u_{tj}(t) = u_j(t) + u_g(t)$$

We start first from a simple MDOF system in which all the degrees of freedom $u_j(t)$ are displacements in the same direction as the ground motion $u_g(t)$ of its support.
The total displacement of each mass m_j is $u_{tj}(t)$; the relative displacement between mass and ground in motion is $u_j(t)$.

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The total displacement of each mass m_j is $u_{tj}(t)$; the relative displacement between mass and ground in motion is $u_j(t)$.
At each time instant, the total and relative displacements of each mass are related by

$$u_{tj}(t) = u_j(t) + u_g(t)$$

which can also be rewritten in vector notation:

$$\mathbf{u}_t(t) = \mathbf{u}(t) + u_g(t)\,\mathbf{j}, \quad \mathbf{j}^T = [1,1,\ldots,1]$$

As we have seen with reference to SDOF systems, since the inertia force acting on each mass is associated to the total acceleration, and hence given by

$$\mathbf{F}_I = \mathbf{M}\ddot{\mathbf{u}}_t(t) = \mathbf{M}\left[\ddot{\mathbf{u}}(t) + \ddot{u}_g(t)\mathbf{j}\right],$$

then the dynamic equilibrium equations, for this case, are written as follows:

$$\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{C}\dot{\mathbf{u}}(t) + \mathbf{K}\mathbf{u}(t) = -\mathbf{M}\mathbf{j}\ddot{u}_g(t)$$

Consider next a generic MDOF system. The total displacement $u_{tj}(t)$, for each degree of freedom, is again the sum of two addends:

$$u_{tj}(t) = u_j(t) + u_{Bj}(t)$$

where now the contribution from the ground motion, $u_{Bj}(t)$, is not just the ground motion $u_g(t)$ itself, because $u_g(t)$ could be in a direction different from that of u_j and u_{tj}.

It is however easy to realize that each component $u_{Bj}(t)$ is simply the static effect of the ground motion – a rigid body component – on the j-th degree of freedom, that can be represented, as a function of the given ground motion $u_g(t)$, as

$$u_{Bj}(t) = i_{gj} u_g(t)$$

in which vector $\mathbf{i}_g$ is called influence vector, and its j-th component i_{gj} represents the contribution to the j-th degree of freedom resulting from the static application of a unit ground displacement $u_g(t) = 1$.

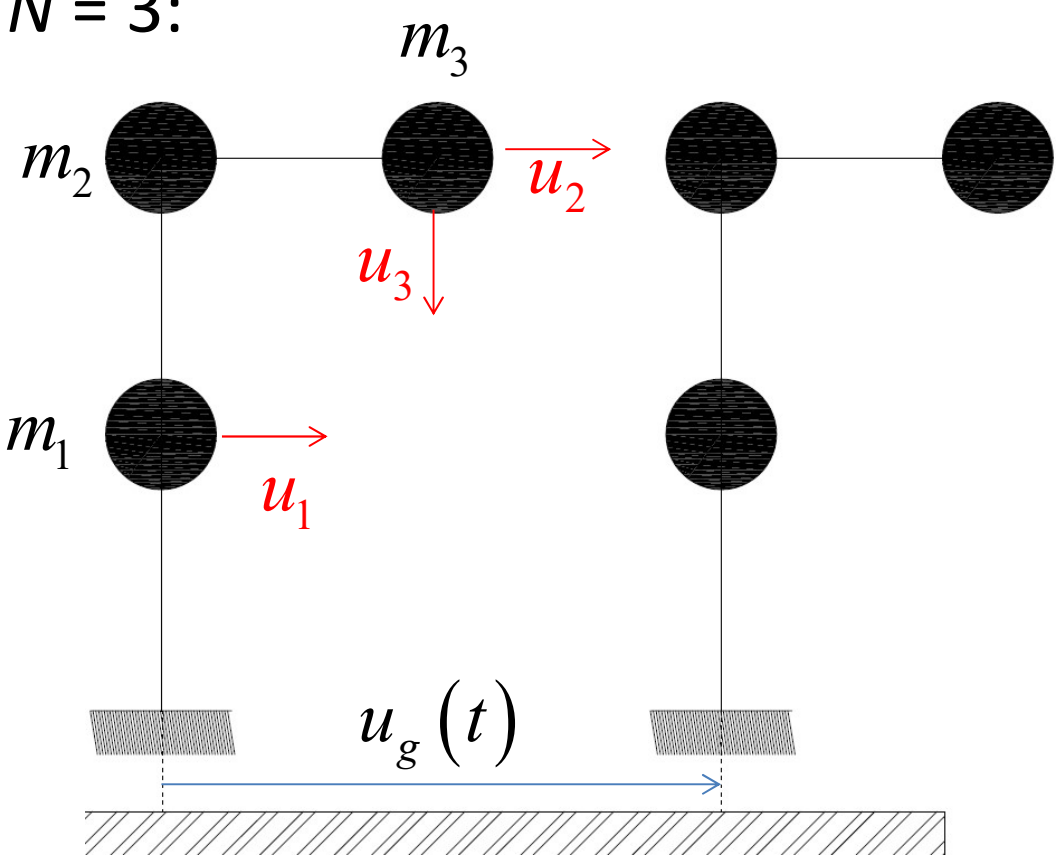
In vector notation, for all the masses of the MDOF system and all the relevant degrees of freedom:

$\mathbf{u}_B(t) = \mathbf{i}_g u_g(t)$ and $\mathbf{u}_t(t) = \mathbf{u}(t) + \mathbf{u}_B(t) = \mathbf{u}(t) + \mathbf{i}_g u_g(t)$

Then, the equations of motion are written as:

$$\mathbf{M}\ddot{\mathbf{u}}(t)+\mathbf{C}\dot{\mathbf{u}}(t)+\mathbf{K}\mathbf{u}(t)=-\mathbf{M}\mathbf{i}_g\ddot{u}_g(t)$$

For example, consider the following simple frame, in which both elements are assumed to be axially rigid, and therefore N = 3:



$$\mathbf{u}(t)=\begin{bmatrix}u_1(t)\\u_2(t)\\u_3(t)\end{bmatrix};\mathbf{i}_g=\begin{bmatrix}1\\1\\0\end{bmatrix};$$

$$\mathbf{M}=\begin{bmatrix}m_1 & 0 & 0\\0 & m_2+m_3 & 0\\0 & 0 & m_3\end{bmatrix}$$

We are now dealing with a generic MDOF system subjected to a generic forcing term, defined by the following vector **F**(*t*):

$$\mathbf{F}(t) = -\mathbf{M}\mathbf{i}_g \ddot{u}_g(t)$$

We then apply the modal transformation to uncouple the governing system of equations:

$$\mathbf{u}(t) = \mathbf{T}_M \mathbf{p}(t);$$

$$\mathbf{p}(t) = \mathbf{T}_M{}^{-1} \mathbf{u}(t);$$

where the transformation matrix $\mathbf{T}_M$ is given by all the natural mode vectors $\psi_{(i)}$:

$$\mathbf{T}_M = \left[\psi_{(1)} \ \psi_{(2)} \ \ldots \ \psi_{(N)} \right]$$

and we always assume that all the eigenvectors $\psi_{(i)}$ are orthonormal with respect to the mass matrix.

This means that the following orthogonality relationships hold:

$$\boldsymbol{\psi}_{(i)}{}^{T}\mathbf{M}\boldsymbol{\psi}_{(j)} = \delta_{ij} \Rightarrow \mathbf{T}_{M}{}^{T}\mathbf{M}\mathbf{T}_{M} = \mathbf{I};$$

$$\boldsymbol{\psi}_{(i)}{}^{T}\mathbf{K}\boldsymbol{\psi}_{(j)} = \omega^{2}{}_{i}\delta_{ij} \Rightarrow \mathbf{T}_{M}{}^{T}\mathbf{K}\mathbf{T}_{M} = \boldsymbol{\Omega};$$

so that new equations of motion become:

$$\ddot{\mathbf{p}}(t) + \widetilde{\mathbf{C}}\dot{\mathbf{p}}(t) + \boldsymbol{\Omega}\mathbf{p}(t) = -\mathbf{T}_{M}{}^{T}\mathbf{M}\mathbf{i}_{g}\ddot{u}_{g}(t)$$

In this form, we assume that also the transformed damping matrix

$$\widetilde{\mathbf{C}} = \mathbf{T}_{M}{}^{T}\mathbf{C}\mathbf{T}_{M}$$

is diagonal, as it happens, for example, if the starting matrix **C** is a Rayleigh damping matrix.

In order to clearly distinguish between the transient and the "static" part of the loading term, we introduce the symbol Γ for the constant vector which multiplies the ground acceleration in the external force vector:

$$\Gamma = \mathbf{T}_M{}^T \mathbf{M}\mathbf{i}_g$$

Each component Γ_n of vector Γ, associated to the corresponding transformed degree of freedom p_n:

$$\Gamma_n = \boldsymbol{\psi}_{(n)}{}^T \mathbf{M}\mathbf{i}_g$$

is called **modal participation factor**, because it can be seen as a measure of the degree to which the *n*-th mode participates to the response to a specific, given ground motion.

Consider now the single *n*-th equation of this uncoupled system, re-written in the following customary form:

$$\ddot{p}_n(t) + 2\zeta_n \omega_n \dot{p}_n(t) + \omega^2 p_n(t) = -\Gamma_n \ddot{u}_g(t)$$

Next, in order to cancel the constant term Γ_n at the r.h.s, use the following change of variable:

$$p_n(t) = -\Gamma_n D_n(t)$$

to obtain the following equivalent form:

$$\ddot{D}_n(t) + 2\zeta_n \omega_n \dot{D}_n(t) + \omega^2 D_n(t) = -\ddot{u}_g(t)$$

This equation is now completely standard, and its solution $D_n(t)$ can be found using the techniques presented with reference to the SDOF systems.

From vector $\mathbf{D}(t)$ vector $\mathbf{p}(t)$ can be immediately computed, and then, recalling that $\mathbf{u}(t) = \mathbf{T}_M\, \mathbf{p}(t)$, one can write vector $\mathbf{u}_{(n)}(t)$, the contribution of the n-th mode to the full displacement vector $\mathbf{u}(t)$, as

$$\mathbf{u}_{(n)}\left(t\right) = \boldsymbol{\Psi}_{(n)} p_n\left(t\right) = \boldsymbol{\Psi}_{(n)} \Gamma_n D_n\left(t\right)$$

where $\Psi_{(n)}$ is the n-th natural mode vector.
Observe that each modal displacement vector $\mathbf{u}_{(n)}(t)$ has the same form as the natural solutions $\mathbf{z}_{(n)}(t)$, studied at the beginning of this part of the class: it is a vector having a constant shape, defined by the eigenvector $y_{(n)}$, and in which all the components follow the same time dependence, defined by the scalar component $p_n(t)$.

By superimposing all the modal contribution vectors one finally computes the full displacement vector **u**(*t*):

$$\mathbf{u}(t) = \mathbf{u}_{(1)}(t) + \mathbf{u}_{(2)}(t) + \ldots + \mathbf{u}_{(N)}(t) =$$
$$= \boldsymbol{\psi}_{(1)} p_1(t) + \boldsymbol{\psi}_{(2)} p_2(t) + \ldots + \boldsymbol{\psi}_{(N)} p_N(t) =$$
$$= \mathbf{T}_M \mathbf{p}(t)$$

Once vector **u**(*t*) is determined, it might be of interest to evaluate the structural response also in terms of other quantities, such as velocities and accelerations, which are trivial, or static quantities.

For example, it may be important, also recalling the meaning of the pseudoaccelerations in the response spectrum analysis, to compute the so-called <u>equivalent static forces</u> $\mathbf{f}_S$ (*t*).

The equivalent static forces, $\mathbf{f}_S(t)$, are the external static forces that produce, at any time t, the displacement $\mathbf{u}(t)$ at the same time t in a structure having the same stiffness properties of the considered one:

$$\mathbf{f}_S(t) = \mathbf{K}\mathbf{u}(t)$$

The equivalent static forces, $\mathbf{f}S(t)$, are the forces arising in the elastic components of the MDOF system that contribute to its stiffness matrix.

It is possible, and easy, to express these forces directly in terms of the pseudo-accelerations considered in the response spectrum analysis. Using the results of the modal analysis, one can write:

$$\mathbf{f}_S(t) = \mathbf{K}\mathbf{u}(t) = \mathbf{K}\mathbf{T}_M\mathbf{p}(t)$$

The contribution given by each single *n*-th mode to the internal static force vector – a vector denoted by $\mathbf{f}_{S(n)}(t)$ – is written as

$$\mathbf{f}_{S(n)}\left(t\right)=\mathbf{K}\mathbf{u}_{(n)}\left(t\right)=\mathbf{K}\boldsymbol{\psi}_{(n)}p_n\left(t\right)$$

Recalling now the usual modal identity:

$$\mathbf{K}\boldsymbol{\psi}_{(n)}=\omega_n^2\mathbf{M}\boldsymbol{\psi}_{(n)}$$

one writes

$$\mathbf{f}_{S(n)}\left(t\right)=\mathbf{K}\mathbf{u}_{(n)}\left(t\right)=\mathbf{K}\boldsymbol{\psi}_{(n)}p_n\left(t\right)=\omega_n{}^2\mathbf{M}\boldsymbol{\psi}_{(n)}p_n\left(t\right)=$$

$$=\omega_n{}^2\mathbf{M}\mathbf{u}_{(n)}\left(t\right)=\mathbf{M}\mathbf{a}_{(n)}\left(t\right)$$

where $\mathbf{a}_{(n)}\left(t\right)=\omega^2{}_n\mathbf{u}_{(n)}\left(t\right)$ is the *n*-th mode contribution to the pseudoacceleration vector

$$\mathbf{a}\left(t\right)=\boldsymbol{\Omega}\mathbf{u}\left(t\right)$$

It is interesting to rephrase this expression in terms of the introduced degrees of freedom $D_n(t)$.
To this purpose, one writes, for each single n-th mode contribution:

$$\mathbf{f}_{S(n)}(t) = \mathbf{K}\mathbf{u}_{(n)}(t) = \mathbf{K}\boldsymbol{\psi}_{(n)} p_n(t) = \mathbf{K}\boldsymbol{\psi}_{(n)} \Gamma_n D_n(t)$$

Using once again the modal identity

$$\mathbf{K}\boldsymbol{\psi}_{(n)} = \omega_n^2 \mathbf{M}\boldsymbol{\psi}_{(n)}$$

one can write what follows:

$$\mathbf{f}_{S(n)}(t) = \mathbf{K}\boldsymbol{\psi}_{(n)} \Gamma_n D_n(t) = \omega_n^2 \mathbf{M}\boldsymbol{\psi}_{(n)} \Gamma_n D_n(t) = \omega^2{}_n \mathbf{s}_{(n)} D_n(t)$$

$$\text{with } \mathbf{s}_{(n)} = \mathbf{M}\boldsymbol{\psi}_{(n)} \Gamma_n$$

Finally, we rewrite this as follows:

$$\mathbf{f}_{S(n)}(t) = \mathbf{s}_{(n)} A_n(t) \text{ with } A_{(n)} = \omega_n^2 D_n(t)$$

$$\boxed{\mathbf{f}_{S(n)}(t) = \mathbf{s}_{(n)} A_n(t)}$$

This expresses the *n*-th mode contribution to the equivalent static force as the product of two interesting terms:

1. the *n*-th mode contribution **s**(*n*) to the spatial distribution of the forcing term due to the ground motion. This is a constant vector, that is computed from the results of an eigenvalue/eigenvector analysis, and from the definition of the influence vector $\mathbf{i}_g$ associated to both the properties of the MDOF system and to the given ground motion;

2. and the pseudo-acceleration time response to the ground motion, $A_n(t)$, of the SDOF system corresponding to the *n*-th mode.

This result, obtained for the equivalent static forces, has been expressed in this apparently involuted form because in this way it can be generalized to any type of response.
In fact, once the equivalent static force vector is computed, mode by mode, any other response vector $\mathbf{r}_{(n)}(t)$, again mode by mode, can be seen as a function of vector $\mathbf{f}_{S(n)}(t)$:

$$\mathbf{r}_{(n)}\left(t\right)=\mathbf{r}_{(n)}\left(\mathbf{f}_{S(n)}\left(t\right)\right)=\mathbf{r}_{(n)}\left(\mathbf{s}_{(n)}A_n\left(t\right)\right)=\mathbf{r}_{(n)}\left(\mathbf{s}_{(n)}\right)A_n\left(t\right)$$

Therefore, the vector $\mathbf{r}_{(n)}(t)$, which represents the n-th mode contribution to any response quantity $\mathbf{r}(t)$ due to a given ground motion, can be formally expressed as follows:

$$\mathbf{r}_{(n)}\left(t\right)=\mathbf{r}_{S(n)}A_n\left(t\right)\ \text{with}\ \mathbf{r}_{S(n)}=\mathbf{r}\left(\mathbf{s}_{(n)}\right)$$

where the symbol $\mathbf{r}_{S(n)}$ represents the quasi-static response $\mathbf{r}_{(n)}$ of the MDOF system due to a forcing term given by $\mathbf{s}_{(n)}$

Using this expression, which splits in a simple way the quasi-static and the dynamic parts of the response of the MDOF system, any response quantity can be written as the superposition of all the *N* mode contributions:

$$\mathbf{r}(t) = \sum_{n=1}^{N} \mathbf{r}_{(n)}(t) = \sum_{n=1}^{N} \mathbf{r}_{S(n)} A_n(t)$$

Recall now the definition of combined elastic design spectrum, a statistically obtained piecewise-linear “summary” of the peak response values of displacements, pseudo-velocities and pseudo-accelerations for SDOF systems, due to a “typical” ground motion at a specific geographic location, as functions of their natural periods and of their damping ratios.

Once a design spectrum is available, it is possible to make good use of it to compute the peak quantities, in the mechanical response of a MDOF system, due to the seismic input that is "summarized" into the design spectrum itself, adjusting the modal superposition technique illustrated earlier. This technique is called response spectrum technique.

Denoting with $\mathbf{r}_{(n)}(t)$ the n-th mode contribution to any generic response term $\mathbf{r}(t)$, its <u>peak value</u> $\mathbf{r}_{(n)0}$, associated to a given design spectrum, can be obtained from the very definition of design spectrum.
In fact, recall that the n -th mode contribution $\mathbf{r}_{(n)}(t)$ can be expressed as:

$$\mathbf{r}_{(n)}\left(t\right)=\mathbf{r}_{S(n)}A_n\left(t\right)$$

The term $An(t)$ indicates the pseudo-acceleration response to the given ground motion of the SDOF system corresponding to the n-th mode.

Then, the design pseudo-acceleration spectrum gives us precisely the peak value of $A_n(t)$, corresponding to the frequency ω_n associated to the n-th natural mode of the MDOF system. We denote with A_{n0} this peak value.

Therefore, the peak value $\mathbf{r}_{(n)0}$ of $\mathbf{r}_{(n)}(t)$ is simply given by

$$\mathbf{r}_{(n)0} = \mathbf{r}_{S(n)} A_{n0}$$

and this peak value $\mathbf{r}_{(n)0}$ is usually referred to as the peak modal response for the response quantity of interest.

In this way the calculation of any peak modal response quantity, for a MDOF system, does not require a complete dynamic analysis anymore: vector $\mathbf{r}_{S(n)}$ derives from a quasi-static calculation with vector $\mathbf{s}_{(n)}$ as a given loading term; and the peak pseudo-acceleration A_{n0} is obtained directly from the design spectrum.

Obviously, the dynamics of the MDOF system is still taken into account, in this technique, because, in order to define the "forcing" vector $\mathbf{s}_{(n)}$, one has to compute all the eigenvalues and eigenvectors of the system.

It remains now the only problem of how to combine all the *N* peak modal responses to determine the global peak value of the desired response, i.e., $\mathbf{r}_0 = \max_t \left\{ \mathbf{r}(t) \right\}$

It should be clear that such a quantity $\mathbf{r}_0$ can not be obtained by simply adding all the *N* modal peak contributions $\mathbf{r}_{(n)0}$, because each of these occurs at a different time, and the global response $\mathbf{r}(t)$ reaches its own peak at a yet different time.

Therefore, approximations should be introduced in order to estimate the peak response from a suitable combination of the peak modal contributions.

A first one consists in assuming that all the modal peaks occur at the same time, and in ignoring the algebraic signs of each modal term.

$$\mathbf{r}_0 \leq \sum_{n=1}^{N} \mathbf{r}_{(n)0}$$

This provides a very rough upper bound to the global peak value.

A second, simple technique provides good results for structures with well-separated natural frequencies: the **square-root-of-sum-of-squares (SRSS) rule**

$$\mathbf{r}_0 \approx \sqrt{\sum_{n=1}^{N} \mathbf{r}^2{}_{(n)0}}$$

A third technique is called the **complete quadratic combination (CQC) rule**, and it can be successfully applied to a wide range of structures:

$$\mathbf{r}_0 \approx \sqrt{\sum_{m=1}^{N} \sum_{n=1}^{N} \rho_{mn} \mathbf{r}_{(m)0} \mathbf{r}_{(n)0}}$$

where the coefficients ρ_{mn} are called **correlation coefficients** between the *m*-th and *n*-th modes.

The correlation coefficients ρ_{mn} take values between 0 and 1, and ρ_{mn} = 1 for *m* = *n*. Thus, it is seen that the CQC rule can be rewritten as follows:

$$\mathbf{r}_0 \approx \sqrt{\sum_{n=1}^{N} \mathbf{r}^2{}_{(n)0} + \underbrace{\sum_{m=1}^{N}\sum_{n=1}^{N} \rho_{mn}\mathbf{r}_{(m)0}\mathbf{r}_{(n)0}}_{m\neq n}}$$

where the first summation is the SRSS combination rule, "corrected" by the second summation.

The correlation coefficients ρ_{mn} can be evaluated according to a proposal by Rosenblueth and Elorduy:

$$\rho_{mn} = \frac{1}{1+\varepsilon_{mn}^2} \text{ with } \varepsilon_{mn} = \frac{\omega_m\sqrt{1-\zeta_m{}^2} - \omega_n\sqrt{1-\zeta_n{}^2}}{\zeta'_m\omega_m + \zeta'_n\omega_n}$$

$$\zeta'_n = \zeta_n + \frac{2}{\omega_n s}$$

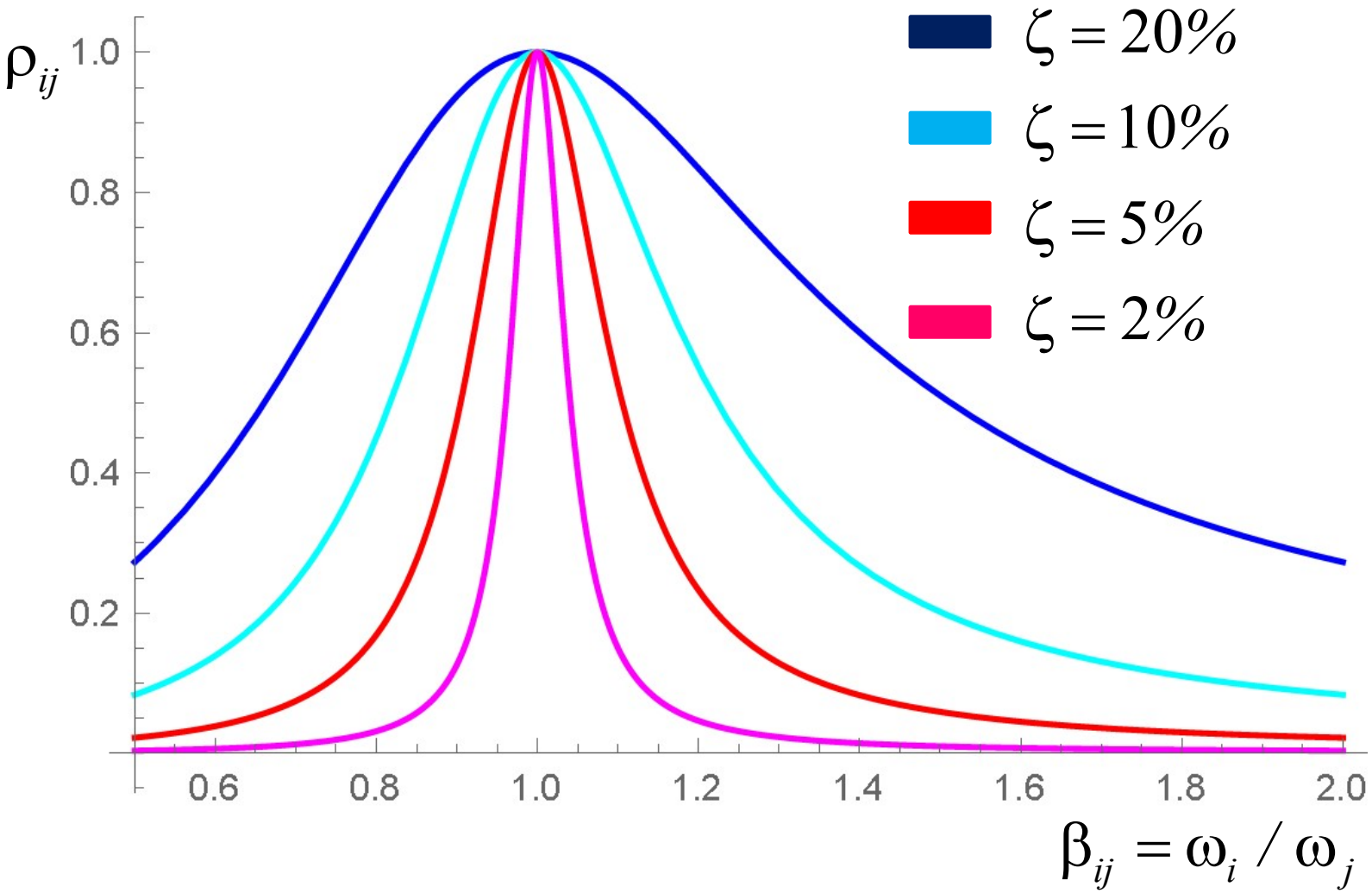
where *s* is the duration of the strong phase of the ground excitation, thought of as an earthquake.

It is instructive to specialize the formula for ρ_{mn} for systems with the same damping ratio ζ in all modes subjected to earthquake excitation with duration s long enough to replace ζ_n' with ζ_n

$$\rho_{mn} = \frac{\zeta^2 \left(1+\beta_{mn}\right)^2}{\left(1-\beta_{mn}\right)^2 + 4\zeta^2\beta_{mn}} \quad \text{where} \quad \beta_{mn} = \frac{\omega_m}{\omega_n}$$

Modal Combination Rule: CQC

correlation coefficient ρ_{ij} for Rosenblueth and Elourdy with damping ratio ζ assumed equal for all the modes



The Rayleigh quotient

Let us finally see, very briefly, a technique for computing the fundamental frequency of a MDOF system in an approximate way.

Recall the eigenvalue problem associated to the calculation of the natural frequencies and modes of a MDOF system with *N* degrees of freedom:

$$\left[\mathbf{K} - \omega^2 \mathbf{M}\right]\boldsymbol{\psi} = \mathbf{0}$$

where both the stiffness matrix **K** and the mass matrix **M** are real and symmetric. The mass matrix is always definite positive, and within the bounds of small-displacement linear elasticity the stiffness matrix can be at worst positive semi-definite for systems that may undergo a rigid body motion.

Consider now the *N* solutions of the eigenvalue/eigenvector problem, i.e., the *N* natural frequencies ω_i and the *N* Eigenvectors $\Psi_{(i)}$, *i* =1, …, *N*. Insert a generic eigenvalue/eigenvector pair in the equation, and premultiply both sides of it by the corresponding eigenvector:

$$\psi_{(i)}{}^T \mathbf{K} \psi_{(i)} = \omega^2{}_i \psi_{(i)}{}^T \mathbf{M} \psi_{(i)}$$

This allows us to express the eigenvalues in terms of the following ratio:

$$\omega^2{}_i = \frac{\psi_{(i)}{}^T \mathbf{K} \psi_{(i)}}{\psi_{(i)}{}^T \mathbf{M} \psi_{(i)}}$$

and to observe that the numerator, in it, is proportional to the strain energy, and the denominator to the kinetic energy, both associated to the *i*-th mode only.

This result is obvious if the eigenvectors are normalized with respect to the mass matrix, but it holds also for any other type of normalization, or even in the absence of normalization.

From a practical point of view, this is of no use, since we can not compute first the eigenvectors and next the eigenvalues of a problem.

But one can define the same ratio for any generic constant displacement vector **u** (a "shape" vector), obtaining a scalar function of this vector enjoying interesting properties:

$$\boxed{R(\mathbf{u}) = \frac{\mathbf{u}^T \mathbf{K} \mathbf{u}}{\mathbf{u}^T \mathbf{M} \mathbf{u}}}$$

$$R(\mathbf{u}) = \frac{\mathbf{u}^T \mathbf{K} \mathbf{u}}{\mathbf{u}^T \mathbf{M} \mathbf{u}}$$

This ratio, a scalar function of vector **u**, is called **Rayleigh quotient**. The Rayleigh quotient can be computed for any choice of a constant displacement vector **u**.

Its most interesting property follows from the study of its first derivative with respect to **u**:

$$\frac{\partial R(\mathbf{u})}{\partial \mathbf{u}} = \frac{2\mathbf{K}\mathbf{u}\left(\mathbf{u}^T \mathbf{M} \mathbf{u}\right) - 2\mathbf{M}\mathbf{u}\left(\mathbf{u}^T \mathbf{K} \mathbf{u}\right)}{\left(\mathbf{u}^T \mathbf{M} \mathbf{u}\right)^2} =$$

$$= \frac{2\mathbf{K}\mathbf{u}}{\mathbf{u}^T \mathbf{M} \mathbf{u}} - 2R(\mathbf{u})\frac{\mathbf{M}\mathbf{u}}{\mathbf{u}^T \mathbf{M} \mathbf{u}} = 2\frac{\mathbf{K}\mathbf{u} - R(\mathbf{u})\mathbf{M}\mathbf{u}}{\mathbf{u}^T \mathbf{M} \mathbf{u}}$$

It is apparent that equating to zero this derivative:

$$\frac{\partial R(\mathbf{u})}{\partial \mathbf{u}} = 0$$

i.e., prescribing the stationarity of $R(\mathbf{u})$ with respect to $\mathbf{u}$, one obtains the starting eigenvalue/eigenvector problem:

$$\left[\mathbf{K} - R(\mathbf{u})\mathbf{M}\right]\mathbf{u} = \mathbf{0}$$

This means that Rayleigh quotient has a stationarity point in correspondence to each eigenvector $\Psi_{(i)}$, and the corresponding value of $R(\mathbf{u})$ is the associated eigenvalue, i.e., the natural frequency ω_i^2.

It is also possible to prove that Rayleigh quotient has an absolute minimum in correspondence to the fundamental frequency/mode couple, i.e., the lowest natural frequency of the system ω_1 and associated eigenvector $\Psi_{(1)}$.

One can thus write what follows:

$$\omega_1^2 = \min_{\mathbf{u}} \left\{ R\left(\mathbf{u}\right) \right\} = \min_{\mathbf{u}} \left\{ \frac{\mathbf{u}^T \mathbf{K} \mathbf{u}}{\mathbf{u}^T \mathbf{M} \mathbf{u}} \right\}$$

This is often called <u>Rayleigh principle</u>.

Rayleigh quotient gives us a quick idea about how the natural frequency of a MDOF system varies with the stiffness and the mass properties for a given displacement vector **u**.

It is in fact easy to see that:

- when the entries in the stiffness matrix increase (stiffer material, more constraints, etc.), the values of the natural frequencies increase (and viceversa);
- when the entries in the mass matrix increase (denser material), the values of the natural frequencies decrease (and viceversa).

Rayleigh quotient can also be used as a simple tool to estimate a natural frequency – most often the fundamental frequency – of a MDOF system.

If one has a reasonable approximation for any eigenvector (as said, in most cases of the first one), say $\tilde{\mathbf{u}}$, then inserting it into Rayleigh quotient one obtains easily an estimate of the corresponding frequency:

$$\omega^2 \approx \frac{\tilde{\mathbf{u}}^T \mathbf{K} \tilde{\mathbf{u}}}{\tilde{\mathbf{u}}^T \mathbf{M} \tilde{\mathbf{u}}}$$